
Linear Independent Component Analysis via Optimal Transport Metric as Contrast*

Abstract

Linear Independent Component Analysis (ICA) recovers mutually independent sources from linear mixtures. Classical algorithms like FastICA, JADE, and InfoMax compute non-Gaussianity via surrogate contrast functions like negentropy approximations, fourth-order cumulants, parametric log-likelihoods, that fail on heterogeneous and high-dimensional mixtures. We propose Optimal Transport metric for ICA, OT-ICA, using the 1D squared Wasserstein distance (W_2^2) to a standard Gaussian as contrast, computed by quantile sorting without density estimation. We prove W_2^2 satisfies a weighted mixture upper bound with strict inequality for non-Gaussian sources, establishing it as a valid ICA contrast. Across $d \in \{10, 20, 30\}$ and five mixture regimes ($N = 10,000$), OT-ICA outperforms FastICA, JADE, InfoMax, and Picard on all continuous and heterogeneous distributions. On purely discrete sources, while W_2^2 retains high contrast resolution, step-function CDFs produce gradient plateaus that hinder convergence. Two applications are presented: EEG ocular artifact removal and structural shock identification in Vector Error Correction Models for financial price discovery.

1 INTRODUCTION

Independent Component Analysis (ICA) [Jutten and Herault, 1985, Comon, 1994] recovers mutually independent source signals $\mathbf{S}$ from a linear mixture $\mathbf{X} = \mathbf{A}\mathbf{S}$. Identifiability holds under the following conditions [Comon, 1994].

Theorem 1 (Linear ICA Identifiability). *Let $\mathbf{X} \in \mathbb{R}^d$ be*

observed data and $\mathbf{S} \in \mathbb{R}^d$ latent sources. Assume (i) $\mathbf{X} = \mathbf{A}\mathbf{S}$ with $\mathbf{A}$ invertible, (ii) $s_1, \dots, s_d$ mutually independent, and (iii) at most one s_i Gaussian. Then $\mathbf{A}$ is identifiable up to column permutation and scaling.

The Central Limit Theorem implies that linear mixtures of independent non-Gaussian sources are more Gaussian than their constituents. ICA recovers sources by maximizing marginal non-Gaussianity over the orthogonal group [Hyvärinen et al., 2001], justified by the following identity.

Theorem 2 (Independence and Non-Gaussianity [Cardoso, 2022]). *Let Y be whitened ($\text{Cov}(Y) = \mathbf{I}$). The mutual information satisfies:*

$$\mathcal{I}(Y) = G(Y) - \sum_i G(Y_i) \quad (1)$$

where $G(\cdot)$ denotes KL divergence to the best-fitting Gaussian. Since $G(Y)$ is invariant under orthogonal rotation, minimizing $\mathcal{I}(Y)$ is equivalent to maximizing $\sum_i G(Y_i)$.

(Proof via KL Pythagorean identities on the Product and Gaussian manifolds: Appendix A.)

Evaluating $G(Y_i)$ requires the marginal density. Classical algorithms replace it with surrogates: FastICA [Hyvärinen et al., 2001, Amari et al., 1996] uses logcosh negentropy approximations, JADE [Cardoso and Souloumiac, 1993] fourth-order cumulants, InfoMax [Bell and Sejnowski, 1995] a parametric log-likelihood, and Picard [Ablin et al., 2018] an adaptive nonlinearity with a provably converging solver. Each proxy fails on specific source geometries (Appendix E).

We propose OT-ICA, using squared Wasserstein metric, W_2^2 to a standard Gaussian as the contrast. Unlike proxy contrasts, Optimal Transport [Monge, 1781, Kantorovich, 1942] provides a true metric on the space of probability distributions, quantifying non-Gaussianity as the cost of transporting the empirical projection to a Gaussian target. The p -Wasserstein distance [Villani, 2003] is the minimum-cost

*Code: https://anonymous.4open.science/r/ot_in_linear_ica-3DBC

coupling:

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int \|x - y\|^p d\pi(x, y) \right)^{1/p} \quad (2)$$

In 1D, Brenier’s theorem [Brenier, 1991] and Caffarelli’s regularity [Caffarelli, 1992] guarantee that the optimal map is monotone, reducing W_2^2 to:

$$W_2^2(\mu, \nu) = \int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|^2 dt \quad (3)$$

This is computable by quantile sorting. Prior work uses 1D W_2 for normality testing [del Barrio et al., 1999] and distribution comparison [Peyré and Cuturi, 2019]; we formalize W_2^2 -to-Gaussian as a valid ICA contrast with finite-sample convergence guarantees.

2 THE OT-ICA FRAMEWORK

After centering and whitening $\mathbf{X}$ to $\tilde{\mathbf{X}}$, the unmixing matrix is constrained to the Orthogonal Group $\mathcal{S}$ [Cardoso, 2022]. OT-ICA finds the projection $\mathbf{b}$ maximizing W_2^2 to the standard Gaussian $\Gamma \equiv \mathcal{N}(0, 1)$:

$$\hat{\mathbf{b}} = \arg \max_{\mathbf{b} \in \mathcal{S}} W_2^2(\mathbb{P}_{\mathbf{b}^\top \tilde{\mathbf{X}}}, \Gamma). \quad (4)$$

The objective is maximized at a pure source direction, established by two bounds.

Theorem 3 (W_2 Upper Bound). *Let $\mathbf{Z} = (Z_1, \dots, Z_d)^\top$ be a vector of mutually independent latent sources, and let $\mathbf{b}$ be a weight vector on the unit sphere ($\sum_{i=1}^d b_i^2 = 1$). The squared Wasserstein distance between the linear mixture $\mathbf{b} \cdot \mathbf{Z}$ and the standard normal distribution Γ is bounded by the weighted sum of the source distances to Γ :*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad (5)$$

Theorem 4 (Strict W_2 Inequality for Non-Gaussian Sources). *Assuming at most one source Z_i is Gaussian, and the source distributions possess smooth and strictly positive densities, the upper bound derived in Theorem 3 is a strict inequality for any valid mixture where multiple weights b_i are non-zero:*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) < \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad (6)$$

The global maximum is attained when $\mathbf{b}$ aligns with a single source direction. Proofs via comonotonicity contradictions under Caffarelli’s regularity theory are in Appendix B.

2.1 OPTIMIZATION ON ORTHOGONAL GROUP $\mathcal{S}$

Three components are required for optimization on the Orthogonal Group $\mathcal{S}$: (i) **Analytical Gaussian Targets** [Fournier and Guillin, 2015], replacing sampled quantile targets with their closed-form bin expectations to eliminate approximation noise; (ii) **Riemannian Gradient Retraction** [Absil et al., 2008], projecting gradients onto the tangent space of $\mathcal{S}$ and retracting via symmetric decorrelation; and (iii) **Gaussian Dithering** [Schuchman, 1964], convolving discrete empirical distributions with a narrow Gaussian kernel to smooth step-function CDFs. Full formulations and pseudocode are in Appendix C.

3 OT-ICA METHODOLOGY VALIDATION

Separation quality is measured by the Amari Performance Index (E) [Amari et al., 1996], the deviation of the gain matrix $\mathbf{P} = \mathbf{B}_{\text{est}} \mathbf{A}$ from a generalized permutation matrix ($E < 0.1$: near-perfect; $E < 0.3$: good separation; $E \geq 0.5$: failure; definition in Appendix D).

Baseline Validation — Amari Error: 0.015

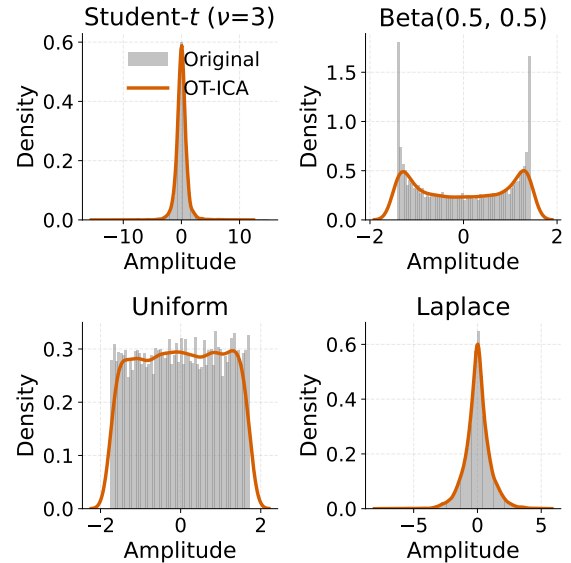


Figure 1: OT-ICA convergence on four standard continuous source types. W_2^2 provides a smooth contrast that drives the Amari error to near zero across optimization iterations.

Figure 1 shows W_2^2 driving the Amari error to near zero on Laplace, Uniform, Student- t , and Beta sources, confirming that the Riemannian solver with analytical Gaussian targets converges reliably on continuous distributions. On this 4D mixture, OT-ICA achieves an Amari error of 0.016 against FastICA’s 0.019; with its global transfer matrix within 10^{-2} of a generalized permutation matrix. (Table 2, Appendix D).

4 EMPIRICAL PERFORMANCE ON HETEROGENEOUS MIXTURES

We compare OT-ICA against FastICA [Hyvärinen et al., 2001], JADE [Cardoso and Souloumiac, 1993], InfoMax [Bell and Sejnowski, 1995], and Picard [Ablin et al., 2018] on five mixture regimes at $d \in \{10, 20, 30\}$, $N = 10,000$, 10 trials per condition (Figure 2).

Continuous and heterogeneous regimes. OT-ICA achieves the lowest Amari error on all four continuous and mixed-type configurations at every dimension (Table 3, Appendix D). On *Continuous Only* sources, OT-ICA reduces error 40–45% over FastICA across all three dimensions. The margin is larger on *Full Hybrid* mixtures (Laplace, Gaussian, Uniform, Student- t , Beta combined): OT-ICA attains $E = 0.046, 0.211, 0.308$ at $d = 10, 20, 30$ against FastICA’s 0.255, 0.451, 0.671—a factor of 2–5 $\times$. On *Zero Gaussian* sources, OT-ICA outperforms InfoMax, which is misspecified for that mixture type, by a similar margin at all dimensions.

Discrete sources. On *Discrete Only* mixtures, OT-ICA un-

derperforms JADE at $d = 10$ (0.643 vs. 0.443), is competitive at $d = 20$ (1.231 vs. JADE 1.267), and at $d = 30$ all proxy-based methods saturate the clip ceiling (1.500) while OT-ICA reaches 1.486. Table 4 shows W_2^2 retains a clear non-Gaussianity signal on count data, exceeding logcosh resolution by more than $10\times$ on standard Poisson($\lambda = 3$), so the failure is not a contrast deficiency. Step-function CDFs create gradient plateaus that stall the Riemannian solver despite the contrast being non-zero (Appendix E.3). Gaussian dithering (Appendix C) is the technique used for avoiding optimization surface discontinuities with discrete components, partially smoothing the CDF steps, but does not fully restore gradient signal at the scales tested.

Method-specific failures. FastICA and Picard [Ablin et al., 2018] produce nearly identical errors across all configurations (within 0.003 on continuous regimes at each dimension), establishing that Picard’s provably converging orthogonal solver does not close the gap: the bottleneck is the shared tanh nonlinearity, not the optimizer. JADE requires $\mathcal{O}(d^4)$ samples for reliable fourth-order cumulant estimation, approximately 810,000 at $d = 30$ against $N = 10,000$ available, yielding $E = 0.489$ on continuous and 0.938 on

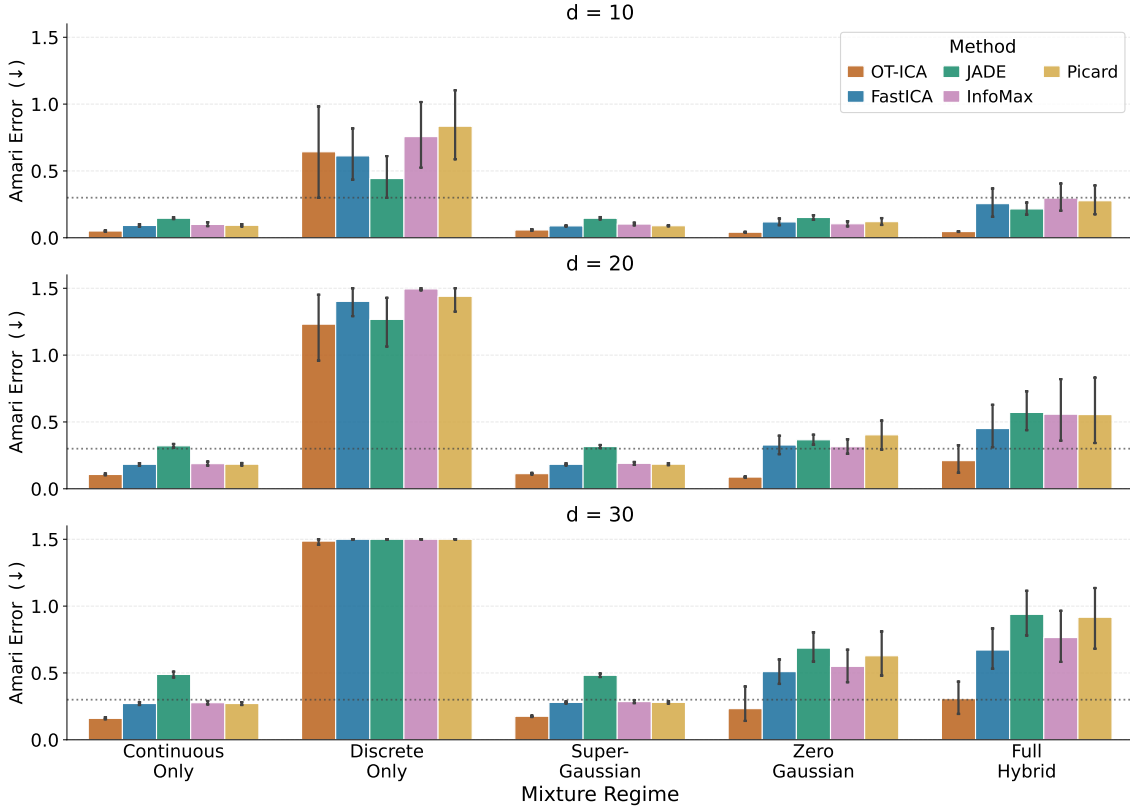


Figure 2: Mean Amari error ($\downarrow$ better, clipped at 1.5) across five mixture regimes and $d \in \{10, 20, 30\}$ ($N = 10,000$, 10 trials, 95% CI). OT-ICA achieves the lowest error on all continuous and heterogeneous regimes at every dimension. On *Discrete Only*, the W_2^2 contrast is present but step-function CDFs produce gradient plateaus that stall the solver (Table 4, Appendix E.3). At $d = 30$, all proxy-based methods saturate the clip ceiling on discrete and several heterogeneous regimes; OT-ICA is the only method below the ceiling on *Discrete Only* ($E = 1.486$). Dotted line at $E = 0.3$ marks the good-separation threshold.

hybrid sources at $d = 30$. InfoMax [Bell and Sejnowski, 1995] is misspecified on distributions mixing super- and sub-Gaussian components, producing $E = 0.765$ on hybrid at $d = 30$ against OT-ICA’s 0.308. (Computational scaling in Appendix F.)

5 INTERDISCIPLINARY APPLICATIONS

5.1 STRUCTURAL IDENTIFICATION IN PRICE DISCOVERY

In fragmented financial markets, cointegrated prices share an unobserved common random walk. The Information Share (IS) [Hasbrouck, 1995, Baillie et al., 2002] quantifies each venue’s contribution, requiring decomposition of the reduced-form residuals u_t of a Vector Error Correction Model (VECM) [Engle and Granger, 1987, Johansen, 1991] into structural shocks ϵ_t via $u_t = B\epsilon_t$.

The covariance constraint $\Omega = BB^\top$ leaves B unidentified up to an orthogonal rotation. Standard Cholesky decomposition resolves this by imposing a recursive market ordering, but IS estimates change with the ordering choice.

Factoring $B = SC$ (whitening times orthogonal rotation) reduces identification to finding C [Zema and Cordon, 2025]. OT-ICA recovers C from the heavy-tailed non-Gaussianity of financial innovations without imposing a recursive ordering (DGP, IS definition, and estimation procedure in Appendix G).

Market / Source	True IS	Estimated IS (Mean)	Std. Dev.
Market 1	0.1200	0.1212	0.0161
Market 2	0.2400	0.2399	0.0245
Market 3	0.6400	0.6389	0.0260

Table 1: Simulation Results for Information Share Recovery across 500 Monte Carlo runs. The estimated IS tightly matches the true IS without imposing arbitrary Cholesky recursive structures.

Table 1 reports mean IS estimates within 0.002 of the true values across 500 Monte Carlo runs, with standard deviations below 0.03, without imposing any market ordering.

5.2 EEG ARTIFACT REMOVAL

The skull acts as a linear volume conductor [Hyvärinen et al., 2001]: ocular blink artifacts mix with neural signals before reaching scalp electrodes. Blink artifacts are impulsive and super-Gaussian; their high W_2^2 separates them from neural components without specifying a density model.

Figure 3 shows OT-ICA concentrating the blink artifact into a single component with excess kurtosis substantially above the remaining four. Zeroing that component and inverting

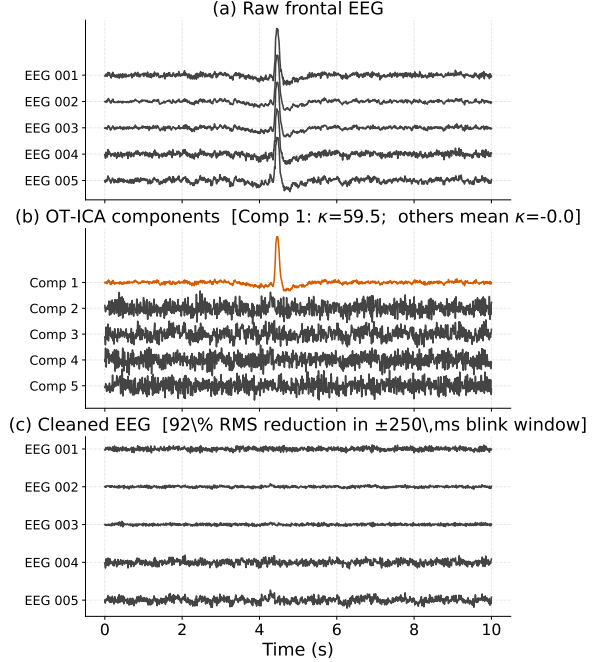


Figure 3: OT-ICA on five frontal EEG channels (MNE sample dataset). Panel (b): isolated blink component (orange) with excess kurtosis κ in the panel title; remaining components have mean κ near zero. Panel (c): signal reconstructed by zeroing the artifact component, with RMS reduction in the ± 250 ms blink window in the panel title.

the total unmixing matrix produces the RMS reduction quantified in the panel title, without tuning a density model.

6 CONCLUSION

OT-ICA replaces proxy contrast functions with the exact W_2^2 metric computed by quantile sorting, avoiding FastICA’s gradient degeneracy, JADE’s $\mathcal{O}(d^4)$ sample complexity, and InfoMax’s distributional misspecification. The Picard comparison isolates the contrast as the failure mode: a faster solver on the same tanh nonlinearity does not reduce Amari error. Across $d \in \{10, 20, 30\}$ and five mixture regimes ($N = 10,000$, 10 trials), OT-ICA achieves the lowest Amari error on all continuous and heterogeneous configurations; on purely discrete sources, gradient plateaus stall the solver, an open optimization problem. OT-ICA incurs higher per-iteration cost than FastICA at large d (Appendix F), but this tradeoff reverses on heterogeneous mixtures where proxy-based methods saturate the error ceiling. Applications to VECM structural identification and EEG artifact removal confirm the method transfers to real data without distributional assumptions. Future work will extend OT-ICA to causal discovery, including as a front-end for LiNGAM [Shimizu et al., 2006], Causal Component Analysis [Liang et al., 2023], and identifiable nonlinear ICA [Buchholz et al., 2022].

References

- Pierre Ablin, Jean-François Cardoso, and Alexandre Gramfort. Faster independent component analysis by preconditioning with Hessian approximations. *IEEE Transactions on Signal Processing*, 66(15):4040–4049, 2018.
- P-A Absil, Robert Mahony, and Rodolphe Sepulchre. *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, 2008.
- Shun-Ichi Amari, Andrzej Cichocki, and Howard H Yang. A new learning algorithm for blind signal separation. In *Advances in neural information processing systems*, pages 757–763, 1996.
- Richard T Baillie, G Geoffrey Booth, Yiuman Tse, and Tatyana Zobotina. Price discovery and common factor models. *Journal of Financial Markets*, 5(3):309–321, 2002.
- Anthony J Bell and Terrence J Sejnowski. An information-maximization approach to blind separation and blind deconvolution. *Neural Computation*, 7(6):1129–1159, 1995.
- Yann Brenier. Polar factorization and monotone rearrangement of vector-valued functions. *Communications on pure and applied mathematics*, 44(4):375–417, 1991.
- Simon Buchholz, Michel Besserve, and Bernhard Schölkopf. Function classes for identifiable nonlinear independent component analysis. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- Luis A Caffarelli. The regularity of mappings with a convex potential. *Journal of the American Mathematical Society*, 5(1):99–104, 1992.
- Jean-François Cardoso. Independent component analysis in the light of information geometry. *Entropy*, 24(3):377, 2022.
- Jean-François Cardoso and Antoine Souloumiac. Blind beamforming for non-Gaussian signals. *IEE Proceedings F – Radar and Signal Processing*, 140(6):362–370, 1993.
- Pierre Comon. Independent component analysis, a new concept? *Signal processing*, 36(3):287–314, 1994.
- Eustasio del Barrio, Juan A Cuesta-Albertos, Carlos Matrán, and Jesús M Rodríguez-Rodríguez. Tests of goodness of fit based on the L_2 -Wasserstein distance. *Annals of Statistics*, 27(4):1230–1239, 1999.
- Robert F Engle and Clive WJ Granger. Co-integration and error correction: Representation, estimation, and testing. *Econometrica: Journal of the Econometric Society*, 55(2): 251–276, 1987.
- Nicolas Fournier and Arnaud Guillin. On the rate of convergence in wasserstein distance of the empirical measure. *Probability Theory and Related Fields*, 162(3):707–738, 2015.
- Joel Hasbrouck. One measure of price discovery in fragmented markets. *The Journal of Finance*, 50(4):1175–1199, 1995.
- Aapo Hyvärinen, Juha Karhunen, and Erkki Oja. *Independent Component Analysis*. John Wiley & Sons, 2001.
- Søren Johansen. Estimation and hypothesis testing of cointegration vectors in gaussian vector autoregressive models. *Econometrica: Journal of the Econometric Society*, 59(6): 1551–1580, 1991.
- Christian Jutten and Jean Herault. Blind separation of sources, part i: An adaptive algorithm based on neuromimetic architecture. *Signal processing*, 24(1):1–10, 1985.
- Leonid V Kantorovich. On the translocation of masses. *C. R. (Doklady) Acad. Sci. URSS (N.S.)*, 37:199–201, 1942.
- Harold W Kuhn. The hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2): 83–97, 1955.
- Te-Won Lee, Mark Girolami, and Terrence J Sejnowski. Independent component analysis using an extended infomax algorithm for mixed subgaussian and supergaussian sources. *Neural Computation*, 11(2):417–441, 1999.
- Wendong Liang, Armin Kekić, Julius von Kügelgen, Simon Buchholz, Michel Besserve, Luigi Gresele, and Bernhard Schölkopf. Causal component analysis. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Gaspard Monge. Mémoire sur la théorie des déblais et des remblais. *Histoire de l’Académie Royale des Sciences de Paris*, 1781.
- Gabriel Peyré and Marco Cuturi. Computational optimal transport: With applications to data science. *Foundations and Trends® in Machine Learning*, 11(5-6):355–607, 2019.
- Leonard Schuchman. Dither signals and their effect on quantization noise. *IEEE Transactions on Communication Technology*, 12(4):162–165, 1964.
- Shohei Shimizu, Patrik O Hoyer, Aapo Hyvärinen, and Antti Kerminen. A linear non-gaussian acyclic model for causal discovery. *Journal of Machine Learning Research*, 7(10), 2006.
- Cédric Villani. *Topics in Optimal Transportation*, volume 58. American Mathematical Society, 2003.

Sebastiano Michele Zema and Francesco Cordonì. A unifying non-gaussian information share approach to price discovery. *Available at SSRN 5234231*, 2025.

Supplementary Material for: Linear Independent Component Analysis via Optimal Transport Metric as Contrast

A INFORMATION GEOMETRY OF ICA

Information geometry provides a framework for quantifying the relationship between independence and non-Gaussianity by analyzing probability distributions as points on geometric manifolds. Following the perspective introduced by Cardoso [2022], ICA can be understood as orthogonally projecting the empirical distribution P_Y onto two distinct subspaces: the Product manifold $\mathcal{P}$ (containing distributions where marginals are independent) and the Gaussian manifold $\mathcal{G}$ (encompassing all multivariate Gaussians).

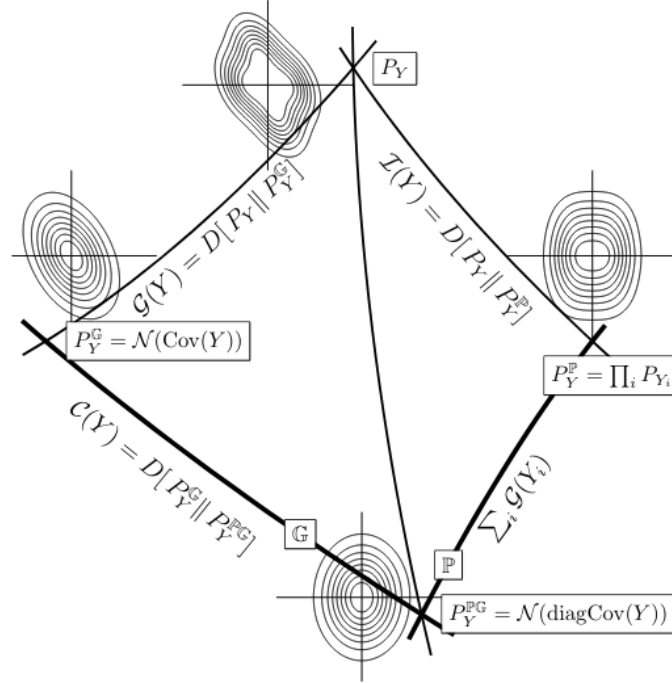


Figure 4: The Pythagorean geometry of Independent Component Analysis. Figure from Cardoso [2022].

The Mutual Information $\mathcal{I}(Y)$ measures the distance from P_Y to its closest independent product distribution P_Y^P . A Pythagorean identity exists on the Product manifold for any candidate product distribution $Q = \prod_i q_i$:

$$\begin{aligned} D_{\text{KL}}(P_Y \| Q) &= \int P_Y(y) \log \left(\frac{P_Y(y) P_Y^P(y)}{P_Y^P(y) Q(y)} \right) dy \\ &= \mathcal{I}(Y) + \sum_i D_{\text{KL}}(P_{Y_i} \| q_i). \end{aligned} \quad (7)$$

A similar Pythagorean identity holds for the Gaussian manifold $\mathcal{G}$. Let $\Phi_Y = \mathcal{N}(0, \text{Cov } Y)$ be the best Gaussian fit (sharing the exact mean and covariance of P_Y), and let $\Phi = \mathcal{N}(0, \Sigma)$ be an arbitrary Gaussian target. The divergence decomposes as:

$$\begin{aligned} D_{\text{KL}}(P_Y \| \Phi) &= \int P_Y(y) \log \left(\frac{P_Y(y) \Phi_Y(y)}{\Phi_Y(y) \Phi(y)} \right) dy \\ &= \int P_Y(y) \log \frac{P_Y(y)}{\Phi_Y(y)} dy + \int P_Y(y) \log \frac{\Phi_Y(y)}{\Phi(y)} dy \\ &= D_{\text{KL}}(P_Y \| \Phi_Y) + \mathbb{E}_{P_Y} [\log \Phi_Y(y) - \log \Phi(y)]. \end{aligned} \quad (8)$$

Because the log-density of a zero-mean Gaussian is a quadratic form, its expectation depends only on the second moments (the covariance matrix) of the variable. Since the true distribution P_Y and its Gaussian fit Φ_Y share the exact same covariance matrix by definition, evaluating the expectation of this quadratic form under P_Y is identical to evaluating it under Φ_Y :

$$\mathbb{E}_{P_Y} \left[\log \frac{\Phi_Y(y)}{\Phi(y)} \right] = \mathbb{E}_{\Phi_Y} \left[\log \frac{\Phi_Y(y)}{\Phi(y)} \right] = D_{\text{KL}}(\Phi_Y \| \Phi). \quad (9)$$

Substituting this equivalence back establishes the Pythagorean identity on the Gaussian manifold:

$$D_{\text{KL}}(P_Y \| \mathcal{N}(\Sigma)) = D_{\text{KL}}(P_Y \| \mathcal{N}(\text{Cov } Y)) + D_{\text{KL}}(\mathcal{N}(\text{Cov } Y) \| \mathcal{N}(\Sigma)). \quad (10)$$

We define the joint non-Gaussianity as $G(Y) = D_{\text{KL}}(P_Y \| \mathcal{N}(\text{Cov } Y))$, and correlation as $C(Y) = D_{\text{KL}}(\mathcal{N}(\text{Cov } Y) \| \mathcal{N}(\text{diag Cov } Y))$. By evaluating the KL divergence from P_Y to a fully factorized Gaussian target $\mathcal{N}(\text{diag Cov } Y)$ via both the Product path and Gaussian path, we yield Cardoso's unified theorem [Cardoso, 2022]:

$$\mathcal{J}(Y) + \sum_i G(Y_i) = G(Y) + C(Y). \quad (11)$$

Crucially, once the data is whitened ($\text{Cov}(Y) = \mathbf{I}$), correlation drops to zero ($C(Y) = 0$). The relationship rearranges as:

$$\mathcal{J}(Y) = G(Y) - \sum_i G(Y_i). \quad (12)$$

Because joint non-Gaussianity $G(Y)$ is invariant under orthogonal rotation, minimizing mutual information (achieving independence) equates to maximizing the sum of marginal non-Gaussianities $\sum G(Y_i)$.

B MATHEMATICAL PROOFS FOR OT-ICA BOUNDS

B.1 PROOF OF THEOREM 3: W_2 UPPER BOUND

Theorem 5 (W_2 Upper Bound). *Let $\mathbf{Z} = (Z_1, \dots, Z_d)^\top$ be a vector of mutually independent latent sources, and let $\mathbf{b}$ be a weight vector on the unit sphere ($\sum_{i=1}^d b_i^2 = 1$). The squared Wasserstein distance between the linear mixture $\mathbf{b} \cdot \mathbf{Z}$ and the standard normal distribution Γ is bounded by the weighted sum of the source distances to Γ :*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad (13)$$

Proof. We construct a specific joint probability distribution, or coupling, using a common source of randomness. Let $\mathbf{N} = (N_1, \dots, N_d)^\top$ be a vector of d independent standard normal variables, $\mathbf{N} \sim \mathcal{N}(0, \mathbf{I}_d)$, assumed to be statistically independent of the sources $\mathbf{Z}$. For each independent latent source Z_i , there exists an optimal transport map T_i that pushes forward the standard normal distribution to the source distribution, denoted as $(T_i)_\# \Gamma = Z_i$.

We construct a random variable X representing the mixture:

$$X = \sum_{i=1}^d b_i T_i(N_i). \quad (14)$$

To compare this mixture to a Gaussian distribution, we construct a target variable Y using the same underlying noise vector $\mathbf{N}$:

$$Y = \sum_{i=1}^d b_i N_i. \quad (15)$$

Given that the weights sum to unity, the resulting variable Y follows the standard normal distribution, $Y \sim \Gamma$. The pair (X, Y) creates a valid coupling. Because W_2^2 is the infimum over all valid couplings, the optimal distance is less than or equal to the cost of our constructed coupling:

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \mathbb{E} [|X - Y|^2] = \mathbb{E} \left[\left(\sum_{i=1}^d b_i (T_i(N_i) - N_i) \right)^2 \right]. \quad (16)$$

Squaring the summation produces diagonal terms ($i = j$) and cross terms ($i \neq j$):

$$|X - Y|^2 = \sum_{i=1}^d b_i^2 (T_i(N_i) - N_i)^2 + \sum_{i \neq j} b_i b_j (T_i(N_i) - N_i)(T_j(N_j) - N_j). \quad (17)$$

Because the underlying noise variables N_i and N_j are independent, and the functions evaluate to a mean of zero, when we take the expectation, all cross terms vanish. We are left with the expectation of the diagonal terms:

$$\mathbb{E}[|X - Y|^2] = \sum_{i=1}^d b_i^2 \mathbb{E}[(T_i(N_i) - N_i)^2]. \quad (18)$$

Recognizing that $\mathbb{E}[(T_i(N_i) - N_i)^2]$ is the definition of the squared Wasserstein distance between the individual source Z_i and Γ , we arrive at the bound:

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad \blacksquare \quad (19)$$

B.2 PROOF OF THEOREM 4: STRICT INEQUALITY

Theorem 6 (Strict W_2 Inequality for Non-Gaussian Sources). *Assuming at most one source Z_i is Gaussian, and the source distributions possess smooth and strictly positive densities, the upper bound derived in Theorem 3 is a strict inequality for any valid mixture where multiple weights b_i are non-zero:*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) < \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad (20)$$

Proof. In a one-dimensional setting, the W_2 distance equals the expected squared difference of a specific coupling if and only if the random variables are comonotonic. By Brenier's Theorem [Brenier, 1991], this requires the gradients to be parallel at every point in the probability space:

$$\nabla X(\mathbf{N}) = \lambda(\mathbf{N}) \nabla Y(\mathbf{N}). \quad (21)$$

Assuming the source distributions possess smooth and strictly positive densities, Caffarelli's regularity theory [Caffarelli, 1992] guarantees that the optimal transport maps T_i are continuously differentiable.

We differentiate both sides to compute their Hessian matrices. On the LHS, the cross-derivatives $\frac{\partial^2 X}{\partial N_i \partial N_j}$ are zero, yielding a diagonal matrix of $b_i T_i''(N_i)$. On the RHS, we differentiate the product $\lambda(\mathbf{N})\mathbf{b}$, yielding a Rank-1 outer product matrix:

$$\underbrace{\begin{bmatrix} b_1 T_1''(N_1) & 0 & \cdots & 0 \\ 0 & b_2 T_2''(N_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b_d T_d''(N_d) \end{bmatrix}}_{\text{Hessian of } X \text{ (Diagonal)}} = \underbrace{\begin{bmatrix} b_1 \frac{\partial \lambda}{\partial N_1} & \cdots & b_1 \frac{\partial \lambda}{\partial N_d} \\ \vdots & \ddots & \vdots \\ b_d \frac{\partial \lambda}{\partial N_1} & \cdots & b_d \frac{\partial \lambda}{\partial N_d} \end{bmatrix}}_{\text{Outer Product } \mathbf{b}(\nabla \lambda)^\top \text{ (Rank-1)}} \quad (22)$$

Examining any off-diagonal entry ($i \neq j$), the LHS dictates it must be zero, establishing $b_i \frac{\partial \lambda}{\partial N_j} = 0$. Because we assume a mixture where multiple weights are non-zero ($b_i \neq 0$), it follows that the gradient of the scaling function is zero ($\nabla \lambda = \mathbf{0}$). Consequently, $\lambda(\mathbf{N})$ is a global scalar constant, λ .

Equating the diagonal elements implies $T_i'(N_i) = \lambda$. Integrating this indicates the optimal transport maps are linear functions of the form:

$$T_i(x) = \lambda x + c. \quad (23)$$

Applying a purely linear transformation to Gaussian noise N_i produces another Gaussian distribution. The identifiability assumption of ICA dictates that the original latent sources Z_i are non-Gaussian. Therefore, the linear map requirement contradicts the non-Gaussianity of the sources. Because the condition for equality cannot be met, the relationship is a strict inequality. $\blacksquare$

B.3 PROOF OF INTRACTABILITY OF NEWTON UPDATES FOR EMPIRICAL OT-ICA

Theorem 7 (Intractability of Newton Updates). *For an empirical distribution mapped to a continuous target via sorting, computing a closed-form Hessian $\mathbf{H} = \nabla_{\mathbf{b}}^2 W_2^2$ is analytically intractable without introducing external density estimation mechanisms. The Map Derivative $\nabla_{\mathbf{b}} T_{\mathbf{b}}$ requires the continuous Probability Density Function (PDF) evaluated at the quantile boundaries.*

Proof. To implement a fixed-point Newton step, the Hessian matrix of the squared Wasserstein distance, $\mathbf{H} = \nabla_{\mathbf{b}}^2 W_2^2$, is required. Let the 1D projection of the data be $y = \mathbf{b}^\top \mathbf{X}$, and let $T_{\mathbf{b}}$ be the optimal transport map pushing this empirical projection to the target Gaussian distribution.

We compute the first differential (the gradient) with respect to $\mathbf{b}$. Utilizing the envelope theorem for optimal transport, the variation of the optimal map evaluates to zero within the cost function:

$$\mathbf{g} = \nabla_{\mathbf{b}} \left(\frac{1}{2} W_2^2 \right) = \mathbb{E} \left[\mathbf{X} (\mathbf{b}^\top \mathbf{X} - T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})) \right]. \quad (24)$$

To compute the Hessian $\mathbf{H}$, we must differentiate the gradient vector $\mathbf{g}$ with respect to $\mathbf{b}$:

$$\mathbf{H} = \nabla_{\mathbf{b}} \mathbf{g} = \mathbb{E} \left[\mathbf{X} \mathbf{X}^\top - \mathbf{X} \nabla_{\mathbf{b}} [T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})]^\top \right]. \quad (25)$$

Applying the total derivative using the chain rule yields two distinct components:

$$\nabla_{\mathbf{b}} [T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})] = \underbrace{T'_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X}) \mathbf{X}}_{\text{Argument Derivative}} + \underbrace{(\nabla_{\mathbf{b}} T_{\mathbf{b}})(\mathbf{b}^\top \mathbf{X})}_{\text{Map Derivative}}. \quad (26)$$

By Caffarelli’s regularity theory [Caffarelli, 1992], the transport map $T_{\mathbf{b}}$ is continuously differentiable, guaranteeing the existence of the Argument Derivative. The fundamental limitation lies within the Map Derivative.

In the 1D optimal transport setting, the transport map relies on the empirical cumulative distribution function, $F_{\mathbf{b}}$, of the projected data y . Expanding the Map Derivative yields:

$$(\nabla_{\mathbf{b}} T_{\mathbf{b}})(y) = \nabla_{\mathbf{b}} [\Phi^{-1}(F_{\mathbf{b}}(y))] = \frac{1}{\phi(\Phi^{-1}(F_{\mathbf{b}}(y)))} \nabla_{\mathbf{b}} F_{\mathbf{b}}(y). \quad (27)$$

The term $\nabla_{\mathbf{b}} F_{\mathbf{b}}(y)$ represents the rate of change of the cumulative probability mass as the projection plane $\mathbf{b}$ is rotated. The derivative of a cumulative distribution function’s boundary requires the underlying continuous probability density function (PDF) evaluated at that specific boundary. For empirical distributions consisting of discrete sampled points, this density does not natively exist, rendering the closed-form calculation of the Hessian analytically intractable. ■

C ALGORITHMIC ENHANCEMENTS AND METHODOLOGY

C.1 WHY W_2^2 INSTEAD OF W_1 ?

While the W_1 distance is robust to outliers, its reliance on an absolute L_1 penalty introduces severe geometric challenges on finite samples.

As shown in Figure 5, the W_1 cost lacks the convexity needed to smooth out empirical noise. Its derivative fluctuates frequently, causing first-order gradient ascent solvers to stall in local non-smooth traps. The squared penalty in the W_2^2 metric acts as a structural smoother, penalizing local distortions and providing a continuous, decelerating gradient profile that allows for reliable convergence.

C.2 ANALYTICAL GAUSSIAN TARGETS

Discrete approximation of the target Gaussian distribution introduces approximation noise. To eliminate this noise, we replace point-sampled targets with an analytical formulation, computing the expected value of a standard normal variable within each discrete quantile bin.

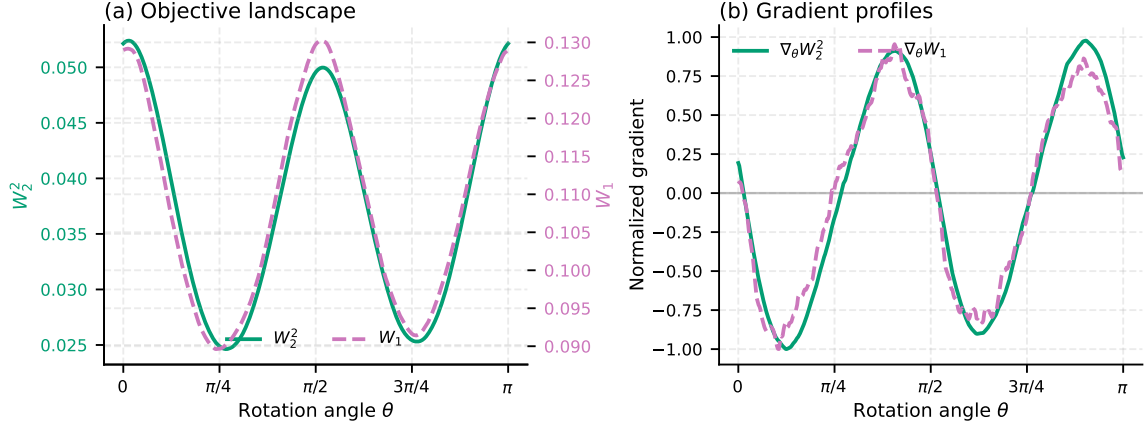


Figure 5: Gradient Landscapes of W_1 and W_2^2 across continuous rotation angles θ . The W_2^2 gradient smoothly decelerates toward zero, whereas the W_1 gradient exhibits jagged volatility on finite empirical samples.

Definition 1 (Analytical Gaussian Target). Let the uniform probability bin edges for N samples be defined as $p_i = \frac{i}{N}$ for $i = 0, \dots, N$, with corresponding Gaussian domain boundaries $z_i = \Phi^{-1}(p_i)$, where Φ is the standard normal CDF. The analytical target value T_i for the i -th sorted sample is:

$$T_i = N \int_{z_{i-1}}^{z_i} x \phi(x) dx = N(\phi(z_{i-1}) - \phi(z_i)). \quad (28)$$

C.3 RIEMANNIAN GRADIENT AND RETRACTION

Because the input data is whitened, the unmixing matrix $\mathbf{B}$ must remain in the Orthogonal Group $\mathcal{S}$, satisfying $\mathbf{B}\mathbf{B}^\top = \mathbf{I}$. Computing the standard Euclidean gradient $\mathbf{G} = \nabla_{\mathbf{B}} W_2^2$ points into the unconstrained ambient space.

Definition 2 (Riemannian Gradient on the Orthogonal Group). Given the Euclidean gradient $\mathbf{G}$, the Riemannian gradient resulting from the projection onto the tangent space $T_{\mathbf{B}}\mathcal{S}$ is given by [Absil et al., 2008]:

$$\nabla_{\mathcal{S}} \mathbf{B} = \mathbf{G} - \frac{1}{2}(\mathbf{G}\mathbf{B}^\top + \mathbf{B}\mathbf{G}^\top)\mathbf{B}. \quad (29)$$

Definition 3 (Symmetric Decorrelation Retraction). To map the matrix back to the orthogonal surface after a tangent step ($\mathbf{B}_{\text{step}}$), we compute the overlap covariance $\mathbf{C} = \mathbf{B}_{\text{step}} \mathbf{B}_{\text{step}}^\top$ and apply the inverse square root:

$$\mathbf{B}_{\text{new}} = \mathbf{C}^{-1/2} \mathbf{B}_{\text{step}}. \quad (30)$$

C.4 CONTINUOUS SMOOTHING VIA GAUSSIAN DITHERING

Applying continuous optimal transport directly to discrete mixtures introduces non-smooth step-functions. Adapted from signal processing [Schuchman, 1964], we inject a variance of continuous, zero-mean Gaussian noise ($\sigma_{\text{dither}}^2 = 0.01$) into the 1D projected components prior to sorting. This decorrelates deterministic quantization errors and convolves the discrete empirical distribution with a Gaussian kernel.

C.5 TOTAL COMPLEXITY AND STATISTICAL EFFICIENCY

For a dataset of dimension d with N samples, K random restarts (batched into a tensor $\mathbf{B}_{\text{batch}}$), and T optimization iterations, the per-iteration complexity is:

$$\mathcal{O}(K \cdot (d \cdot N \log N + d^3)) \quad (31)$$

The $N \log N$ term represents the sorting of projections, while the d^3 term represents the cost of the symmetric decorrelation used for retraction.

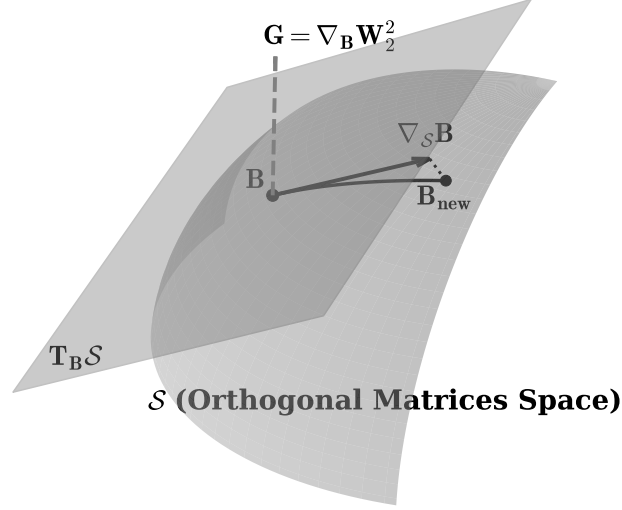


Figure 6: Geometric sketch of optimization on the Orthogonal Group $\mathcal{S}$. The Euclidean gradient $\mathbf{G}$ is projected onto the tangent space to yield the Riemannian gradient $\nabla_{\mathcal{S}} \mathbf{B}$. A retraction maps the estimate back to the orthogonal surface.

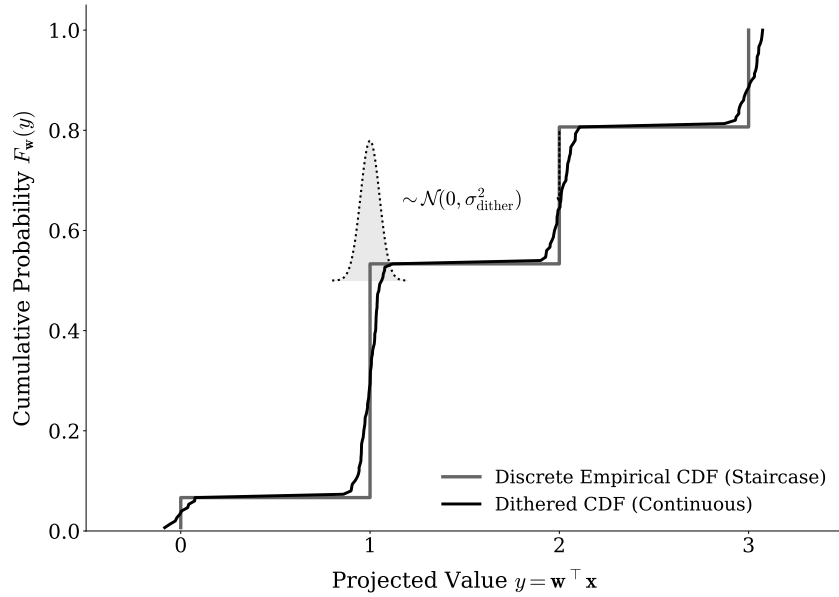


Figure 7: The non-differentiable staircase of a discrete empirical CDF (gray) is convolved with a narrow Gaussian kernel via dithering, yielding a continuous, differentiable curve (black).

Algorithm 1 The Optimal Transport ICA (OT-ICA) Algorithm

Require: Observed mixture $\mathbf{X} \in \mathbb{R}^{d \times N}$, iterations K , learning rate η , batch size N_{batch} , dither noise σ_{dither}

Ensure: Estimated unmixing matrix $\mathbf{B}_{\text{final}} \in \mathbb{R}^{d \times d}$

```
1: Phase 1: Preprocessing & Target Generation
2: Center the data:  $\mathbf{X} \leftarrow \mathbf{X} - \mathbb{E}[\mathbf{X}]$ 
3: Whiten the data:  $\tilde{\mathbf{X}} \leftarrow (\mathbf{X}\mathbf{X}^\top)^{-1/2}\mathbf{X}$ 
4: Compute analytical target  $\mathbf{T} \in \mathbb{R}^{N_{\text{batch}}}$  via analytical Gaussian CDF integration
5: Phase 2: Initialization
6: Initialize  $\mathbf{B} \in \mathbb{R}^{d \times d}$  randomly from  $\mathcal{N}(0, 1)$ 
7: Apply symmetric decorrelation:  $\mathbf{B} \leftarrow (\mathbf{B}\mathbf{B}^\top)^{-1/2}\mathbf{B}$ 
8: Phase 3: Stochastic Riemannian Optimization
9: for  $k = 1$  to  $K$  do
10:   Sample stochastic mini-batch  $\tilde{\mathbf{X}}_{\text{batch}} \in \mathbb{R}^{d \times N_{\text{batch}}}$  from  $\tilde{\mathbf{X}}$ 
11:   Project data onto candidates:  $\mathbf{Y} \leftarrow \mathbf{B}\tilde{\mathbf{X}}_{\text{batch}}$ 
12:   Apply Gaussian Dithering:  $\mathbf{Y}_{\text{dither}} \leftarrow \mathbf{Y} + \mathcal{N}(0, \sigma_{\text{dither}}^2)$ 
13:   Sort each row of  $\mathbf{Y}_{\text{dither}}$  ascending to yield empirical quantiles  $\mathbf{Y}_{\text{sorted}}$ 
14:   Compute Wasserstein objective:  $\mathcal{L} = \frac{1}{N_{\text{batch}}} \sum_{i=1}^d \|\mathbf{Y}_{\text{sorted}}^{(i)} - \mathbf{T}\|_2^2$ 
15:   Backpropagate to compute Euclidean gradient:  $\mathbf{G} = \nabla_{\mathbf{B}} \mathcal{L}$ 
16:   Project to tangent space:  $\nabla_{\mathcal{S}} \mathbf{B} = \mathbf{G} - \frac{1}{2}(\mathbf{G}\mathbf{B}^\top + \mathbf{B}\mathbf{G}^\top)\mathbf{B}$ 
17:   Update matrix along tangent plane:  $\mathbf{B}_{\text{step}} = \mathbf{B} + \eta \nabla_{\mathcal{S}} \mathbf{B}$ 
18:   Retract to Orthogonal Group  $\mathcal{S}$ :  $\mathbf{B} \leftarrow (\mathbf{B}_{\text{step}}\mathbf{B}_{\text{step}}^\top)^{-1/2}\mathbf{B}_{\text{step}}$ 
19: end for
20: Phase 4: Finalization
21:  $\mathbf{B}_{\text{final}} \leftarrow \mathbf{B}(\mathbf{X}\mathbf{X}^\top)^{-1/2}$ 
22: return  $\mathbf{B}_{\text{final}}$ 
```

C.6 THE OT-ICA ALGORITHM

D EXPERIMENTAL METHODOLOGY AND METRICS

D.1 THE GENERALIZED PERMUTATION MATRIX AND AMARI INDEX

To empirically evaluate the success of source separation, we quantify the distance between the estimated unmixing matrix $\mathbf{B}_{\text{est}}$ and the true mixing matrix $\mathbf{A}$. Because ICA is subject to scaling and permutation ambiguities, the target recovery is a generalized permutation matrix. Let $\mathbf{P} = \mathbf{B}_{\text{est}}\mathbf{A}$ be the global transfer matrix. For a $d \times d$ matrix $\mathbf{P}$ with elements p_{ij} , the Amari Performance Index [Amari et al., 1996] measures the structural divergence:

$$E(\mathbf{P}) = \frac{1}{2d} \sum_{i=1}^d \left(\sum_{j=1}^d \frac{|p_{ij}|}{\max_k |p_{ik}|} - 1 \right) + \frac{1}{2d} \sum_{j=1}^d \left(\sum_{i=1}^d \frac{|p_{ij}|}{\max_k |p_{kj}|} - 1 \right). \quad (32)$$

D.2 COMPUTATIONAL RESOURCE REGIMES

Because OT-ICA utilizes stochastic gradient descent while FastICA employs fixed-point Newton iterations, we standardize resources via two defined regimes to ensure fairness:

- **Low Compute Baseline:** OT-ICA is allocated restarts $\min(4 \times d, 150)$ and 150/200 deflation/symmetric phase iterations. FastICA is allocated 50 random restarts and 1,000 maximum iterations per component.
- **High Compute Regime:** OT-ICA is allocated restarts $\min(15 \times d, 600)$ and 300/500 iterations. FastICA is allocated 50 restarts and 5,000 maximum iterations per component.

D.3 METHODOLOGY VALIDATION: GLOBAL TRANSFER MATRICES

We validated OT-ICA on a 4D mixture of Laplace, Uniform, Student- $t(3)$, and Beta(0.5, 0.5) sources with $N = 10,000$, mixed by the random matrix $\mathbf{A}$ below. The global transfer matrices $\mathbf{P} = \hat{\mathbf{B}}\mathbf{A}$ for both algorithms are near-permutation, with Amari errors below 0.02.

True Mixing Matrix A							
$\begin{bmatrix} 1.058 & 1.520 & -0.249 & 1.012 \\ 0.042 & -0.486 & 0.388 & -0.523 \\ -0.154 & -1.572 & -0.304 & -1.234 \\ -0.006 & 0.503 & 0.053 & 0.928 \end{bmatrix}$							
OT-ICA Transfer Matrix P				FastICA Transfer Matrix P			
$\begin{bmatrix} -0.012 & -0.009 & -1.000 & -0.003 \\ 0.000 & 0.003 & -0.001 & 1.000 \\ -0.007 & 1.000 & 0.000 & 0.012 \\ -1.000 & -0.007 & 0.005 & -0.010 \end{bmatrix}$				$\begin{bmatrix} -1.000 & -0.012 & 0.003 & -0.008 \\ 0.012 & -1.000 & 0.001 & -0.004 \\ -0.002 & -0.010 & -0.001 & -1.000 \\ -0.010 & -0.010 & -1.000 & -0.001 \end{bmatrix}$			
Amari Error: OT-ICA = 0.0155 FastICA = 0.0188							

Table 2: Global transfer matrices for the 4D validation mixture. Off-diagonal entries are ≤ 0.012 in absolute value for both algorithms, confirming near-perfect source recovery.

D.4 FULL BENCHMARK RESULTS

Table 3 reports mean Amari errors for all five methods across all mixture regimes at $d \in \{10, 20, 30\}$ ($N = 10,000$, 10 trials). Values at 1.500 indicate total separation failure (clip ceiling). **Bold** marks the lowest error per row.

E LIMITATIONS OF PROXY CONTRAST FUNCTIONS

FastICA approximates negentropy using a non-quadratic contrast function $G(\cdot)$, such as the *logcosh* function. We identify two specific mathematical limitations of this approach.

E.1 THE ZERO NEGENTROPY CONDITION

The approximation isolates non-Gaussianity via the difference of expectations: $\mathbb{E}[G(x)] - \mathbb{E}[G(\nu)]$, where $\nu \sim \mathcal{N}(0, 1)$. If a latent independent source possesses a non-Gaussian distribution such that its expected value under the contrast function equals that of a Gaussian, the approximated negentropy evaluates to zero. The objective function provides no gradient signal, and the algorithm fails to extract the source.

E.2 THE VANISHING CURVATURE CONDITION

FastICA utilizes a Newton (second-order) fixed-point iteration. Defining $g(x) = G'(x)$ and $g'(x) = G''(x)$, FastICA approximates the Hessian matrix dynamically. For a given component i , the diagonal entry scales according to the expectation $\mathbf{H}_{ii} \approx \mathbb{E}[s_i g(s_i) - g'(s_i)]$.

The Newton update requires applying the inverse of this Hessian ($\mathbf{H}^{-1}$). If a non-Gaussian source distribution causes the expectation in the denominator to evaluate to zero, the inverse becomes undefined. This analytical divide-by-zero error makes the update step intractable.

E.3 DISCRETE DISTRIBUTIONS: CONTRAST VS. OPTIMIZATION

The *Discrete Only* failure is mechanistically distinct from the continuous failure modes above. Table 4 compares the non-Gaussianity resolution of W_2^2 against the logcosh proxy negentropy $(\mathbb{E}[G(x)] - \mathbb{E}[G(\nu)])^2$ across standard and highly

Regime	d	OT-ICA	FastICA	JADE	InfoMax	Picard
Continuous Only	10	0.051	0.092	0.146	0.100	0.093
	20	0.107	0.183	0.321	0.188	0.184
	30	0.160	0.271	0.489	0.277	0.271
Strictly Super-Gaussian	10	0.058	0.089	0.146	0.103	0.090
	20	0.113	0.183	0.316	0.190	0.184
	30	0.176	0.280	0.482	0.286	0.280
Zero Gaussian	10	0.042	0.118	0.152	0.105	0.121
	20	0.088	0.327	0.366	0.315	0.403
	30	0.233	0.510	0.686	0.549	0.628
Full Hybrid	10	0.046	0.255	0.216	0.297	0.277
	20	0.211	0.451	0.571	0.558	0.555
	30	0.308	0.671	0.938	0.765	0.917
Discrete Only	10	0.643	0.613	0.443	0.757	0.835
	20	1.231	1.402	1.267	1.495	1.440
	30	1.486	1.500	1.500	1.500	1.500
Pure Laplace	10	—	—	—	—	—
	20	—	—	—	—	—
	30	—	—	—	—	—
Pure Uniform	10	—	—	—	—	—
	20	—	—	—	—	—
	30	—	—	—	—	—
Pure Student- t	10	—	—	—	—	—
	20	—	—	—	—	—
	30	—	—	—	—	—
Pure Bernoulli	10	—	—	—	—	—
	20	—	—	—	—	—
	30	—	—	—	—	—
Pure Poisson	10	—	—	—	—	—
	20	—	—	—	—	—
	30	—	—	—	—	—
Pure Binomial	10	—	—	—	—	—
	20	—	—	—	—	—
	30	—	—	—	—	—
Pure Chi-square	10	—	—	—	—	—
	20	—	—	—	—	—
	30	—	—	—	—	—
Pure Exponential	10	—	—	—	—	—
	20	—	—	—	—	—
	30	—	—	—	—	—

Table 3: Mean Amari error ($\downarrow$ better, clipped at 1.500) for all five methods and mixture regimes ($N = 10,000$, 10 trials). OT-ICA achieves the lowest error on all continuous and heterogeneous configurations. On Discrete Only, JADE leads at $d = 10$; OT-ICA leads at $d = 20$ and is the only method below the clip ceiling at $d = 30$. Pure single-distribution rows filled after running `tpm2026_figures.py` Cell 8b.

non-Gaussian parameterizations.

Distribution	W_2^2	Logcosh Negentropy
Laplace (continuous baseline)	0.0398	0.001214
Binomial (standard, $n = 10$)	0.0344	0.000031
Poisson (standard, $\lambda = 3.0$)	0.0513	≈ 0
Binomial (non-Gaussian, $n = 2$)	0.2022	0.000267
Poisson (non-Gaussian, $\lambda = 0.5$)	0.3384	0.000085

Table 4: W_2^2 provides $\gg 10\times$ greater non-Gaussianity resolution than logcosh on count-based data. Standard Poisson ($\lambda = 3.0$) evaluates to near-zero under logcosh yet retains a clear W_2^2 signal. The failure to unmix discrete sources therefore cannot originate from an insufficient W_2^2 contrast; the issue is in the optimization landscape.

Discrete Mixture Optimization Landscape: OT-ICA vs. FastICA

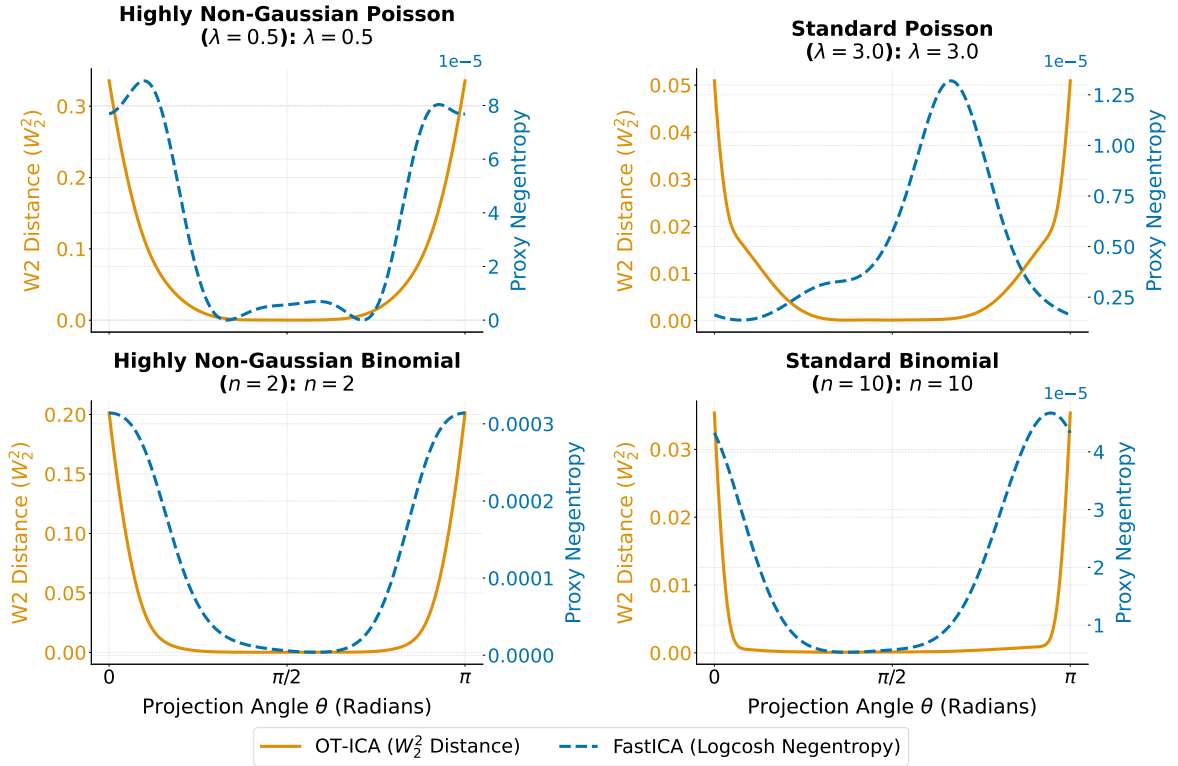


Figure 8: W_2^2 and logcosh optimization landscapes for two-source Poisson and Binomial mixtures across parameterizations. The W_2^2 landscape retains a non-zero signal throughout (confirming the contrast is present), but the step-function geometry of discrete CDFs creates flat plateaus where the gradient evaluates to near-zero, stalling gradient-based solvers. The logcosh landscape degenerates to near-zero for standard count-based parameterizations, indicating a fundamental contrast failure.

E.4 JADE: SAMPLE COMPLEXITY OF FOURTH-ORDER CUMULANTS

JADE [Cardoso and Souloumiac, 1993] identifies sources by jointly diagonalizing a set of fourth-order cumulant matrices via Jacobi sweeps. Reliable estimation of the full cumulant tensor requires $\mathcal{O}(d^4)$ samples [Cardoso and Souloumiac, 1993, Hyvärinen et al., 2001]. At $d = 30$, this implies approximately 810,000 samples; against the $N = 10,000$ available in our benchmark, the cumulant matrices are heavily undersampled. The resulting joint diagonalization converges on estimated tensors reflecting sampling noise rather than true source structure, yielding $E = 0.489$ on continuous sources and $E = 0.938$ on hybrid sources at $d = 30$ —errors that approach or saturate the failure threshold despite JADE’s theoretical consistency

on well-sampled data.

E.5 INFOMAX: SCORE FUNCTION MISSPECIFICATION

InfoMax [Bell and Sejnowski, 1995] maximizes a log-likelihood under a fixed logistic nonlinearity, implicitly treating all sources as sub-Gaussian. On mixtures containing super-Gaussian components (Laplace, Student- t), the score function is misspecified: the gradient points away from the true unmixing direction. Lee et al. [1999] documented this failure mode explicitly, proposing extended InfoMax with online source-type switching. That approach partially addresses the misspecification but requires correct per-component classification, an assumption OT-ICA avoids entirely. In our benchmark, InfoMax produces $E = 0.765$ on hybrid sources and $E = 0.549$ on Zero Gaussian sources at $d = 30$, against OT-ICA’s 0.308 and 0.233 respectively, consistent with the misspecified-score explanation.

E.6 PICARD: THE CONTRAST AS THE BINDING CONSTRAINT

Picard [Ablin et al., 2018] replaces InfoMax’s gradient ascent with an L-BFGS preconditioned solver on the same log-likelihood objective, achieving provably faster convergence per iteration. In our benchmark, FastICA and Picard produce Amari errors within 0.003 of each other across all continuous configurations at each dimension. This near-identical performance isolates the source of failure: solver speed is not the binding constraint. The shared tanh nonlinearity, and not the optimizer, determines the error floor. Replacing it with an exact, distribution-free contrast reduces error by 40–45% on continuous sources.

F COMPUTATIONAL SCALING AND DIMENSIONALITY LIMITS

We benchmarked OT-ICA against FastICA using a linear mixture of continuous Laplace sources across increasing dimensions with a fixed sample size of $N = 10,000$.

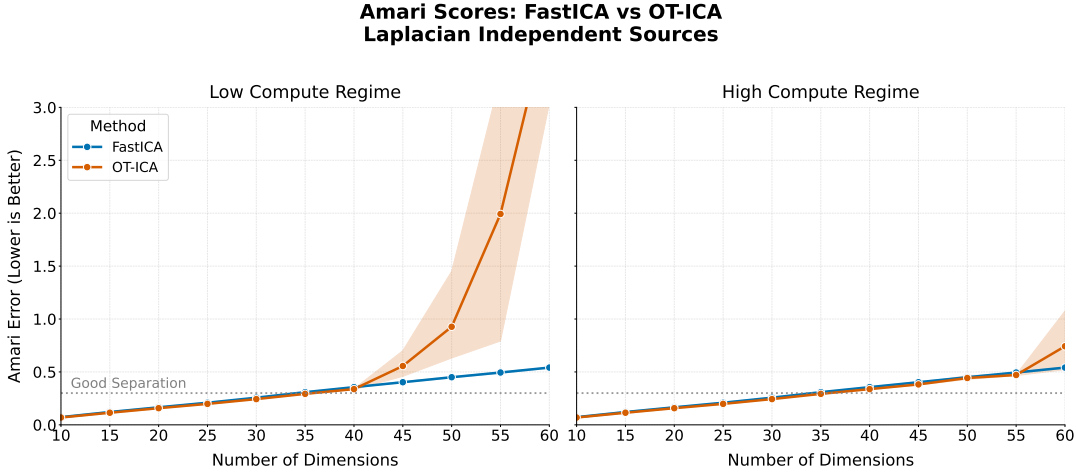


Figure 9: Amari error across dimensions for Laplacian sources at fixed $N = 10,000$ samples. Both OT-ICA and FastICA hit the same curse-of-dimensionality noise ceiling at $d = 40$ —an artifact of finite-sample sparsity shared by all empirical ICA methods at this regime, not a deficiency specific to OT-ICA.

While OT-ICA maintains error rates comparable to FastICA ($E < 0.3$) in lower dimensions, both algorithms exhibit a loss in unmixing quality at $d = 40$. This limit is a manifestation of the curse of dimensionality. The expected error of the empirical Wasserstein distance scales as $\mathcal{O}(N^{-1/d})$. As d grows, a fixed finite sample fails to densely populate the state space, creating massive empty regions that swallow the gradient signal.

The execution time for OT-ICA scales more steeply than FastICA. FastICA evaluates to $\mathcal{O}(dN)$ per iteration, making it computationally superior for exclusively homogeneous, continuous mixtures that do not trigger its proxy blind spots. OT-ICA incorporates parallel sorting across multiple restarts ($\mathcal{O}(K \cdot dN \log N)$), trading execution speed for reliability on complex topologies.

Time Complexity: FastICA vs OT-ICA Laplacian Independent Sources

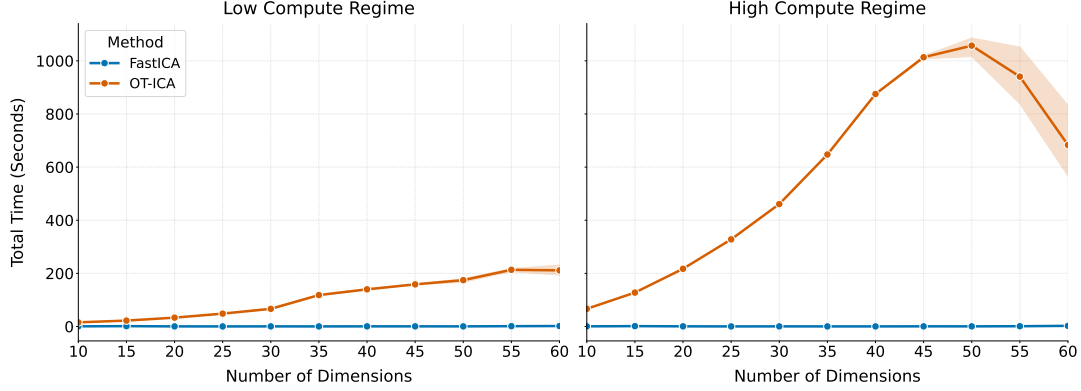


Figure 10: Total execution time (in seconds) for FastICA and OT-ICA across dimensions.

G PRICE DISCOVERY APPLICATION: DATA GENERATING PROCESS AND IS DEFINITION

G.1 DATA GENERATING PROCESS

We simulate a three-market VECM following the non-Gaussian information-share framework of Zema and Cordoni [2025]. The three markets share a single common efficient price, with long-run impact vector $\psi = (1, 1, 1)^\top$ (each market moves one-for-one with the common trend). The true structural mixing matrix is diagonal:

$$\mathbf{B}_{\text{true}} = \text{diag}(\sqrt{0.12}, \sqrt{0.24}, \sqrt{0.64}), \quad (33)$$

which, as shown below, yields the target Information Shares $[0.12, 0.24, 0.64]$ by construction.

Structural innovations ϵ_t are drawn from Student- t distributions with degrees of freedom $\nu_1 = 5$, $\nu_2 = 6$, $\nu_3 = 7$, normalized to unit variance. The VECM reduced-form residuals are $u_t = \mathbf{B}_{\text{true}} \epsilon_t$. Heavy tails are a realistic feature of high-frequency financial innovations and are what enables ICA identification: the non-Gaussianity of u_t carries the rotation information that a Gaussian model would lose.

G.2 INFORMATION SHARE DEFINITION

Under Hasbrouck [1995], the Information Share of venue i measures its contribution to price discovery:

$$\text{IS}_i = \frac{(\psi^\top \mathbf{B})_i^2}{\psi^\top \mathbf{B} \mathbf{B}^\top \psi}. \quad (34)$$

With $\psi = (1, 1, 1)^\top$ and diagonal $\mathbf{B}_{\text{true}}$, the numerator equals B_{ii}^2 and the denominator equals $\sum_j B_{jj}^2 = 0.12 + 0.24 + 0.64 = 1$, so $\text{IS}_i = B_{ii}^2 = \{0.12, 0.24, 0.64\}$.

G.3 ESTIMATION PROCEDURE

The reduced-form covariance is $\mathbf{\Omega} = \mathbf{B} \mathbf{B}^\top$. Cholesky or eigendecomposition of $\mathbf{\Omega}$ yields the whitening transform $\mathbf{S} = \mathbf{\Omega}^{1/2}$; the orthogonal rotation $\mathbf{C}$ satisfying $\mathbf{B} = \mathbf{S} \mathbf{C}$ is then identified from the non-Gaussianity of $\mathbf{S}^{-1} u_t$ by OT-ICA. In this simulation $\mathbf{\Omega}$ is treated as known; in practice it is estimated from VECM residuals by OLS. Two economic constraints are imposed post-estimation: the Hungarian algorithm [Kuhn, 1955] resolves the permutation ambiguity by assigning each structural shock to the market for which it explains maximum variance; sign normalization enforces a positive own-market impact. IS estimates are then computed from $\hat{\mathbf{B}} = \mathbf{S} \hat{\mathbf{C}}$ using the Hasbrouck formula above.